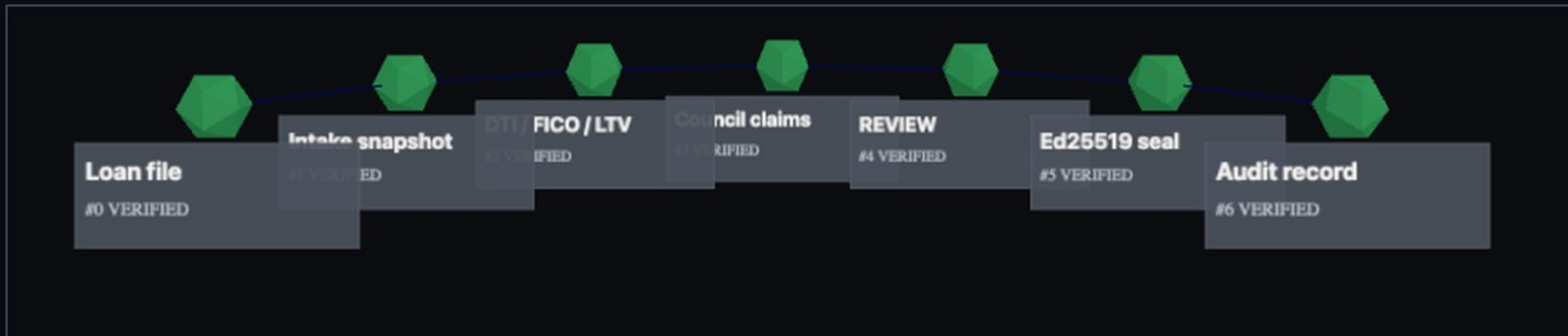


Shadow

A Cryptographically Verifiable Evidence and Spatial
Audit System for AI-Assisted Decisions

From AI answers to independently verifiable decision evidence.



Real render — Shadow 3D audit arc · research prototype

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Practice presentation
2026-07-21

An answer is not an evidence record

AI ANSWER

“Approve the loan.”

fluent · unverifiable



EVIDENCE RECORD

source → tool action → sequence → signature → verifier

structured · signed · independently checkable

SOURCE

What did it rely on?

ACTION

What did it do?

INTEGRITY

Was it changed after?

INDEPENDENCE

Can someone else verify it?

Ordinary logs are mutable and platform-specific; a model's explanation is generated prose, not a record.

From more opinions to verifiable evidence

ORALLEXA

Multiple AI perspectives debate an answer



THE LESSON

More voices do not create verifiability



SHADOW

Portable evidence + independent verification

Trust moved **down a layer** — from the voices to the evidence beneath them. A refinement, not a failure.

A cryptographically bound sequence



Source

doc

—



Action

tool

—



Event

seq 2 · a3f9...

—



Hash chain

commits prev

—



Ed25519 seal

batch root

—



Verify

independent

Change one earlier event → the **first broken link** and its downstream impact become visible.

VERIFIES OFFLINE

SHA-256 + Ed25519

One core, three profiles, many surfaces

EXPERIENCES

CLI · MCP (11) · HTTP · verify.html · Unity Shadow Lens · Three.js replay · 3 explainers

MIXED MATURITY

PROFILES

banking-v1 · data-science-v1 · coding-agent-v1

ONE GRAMMAR

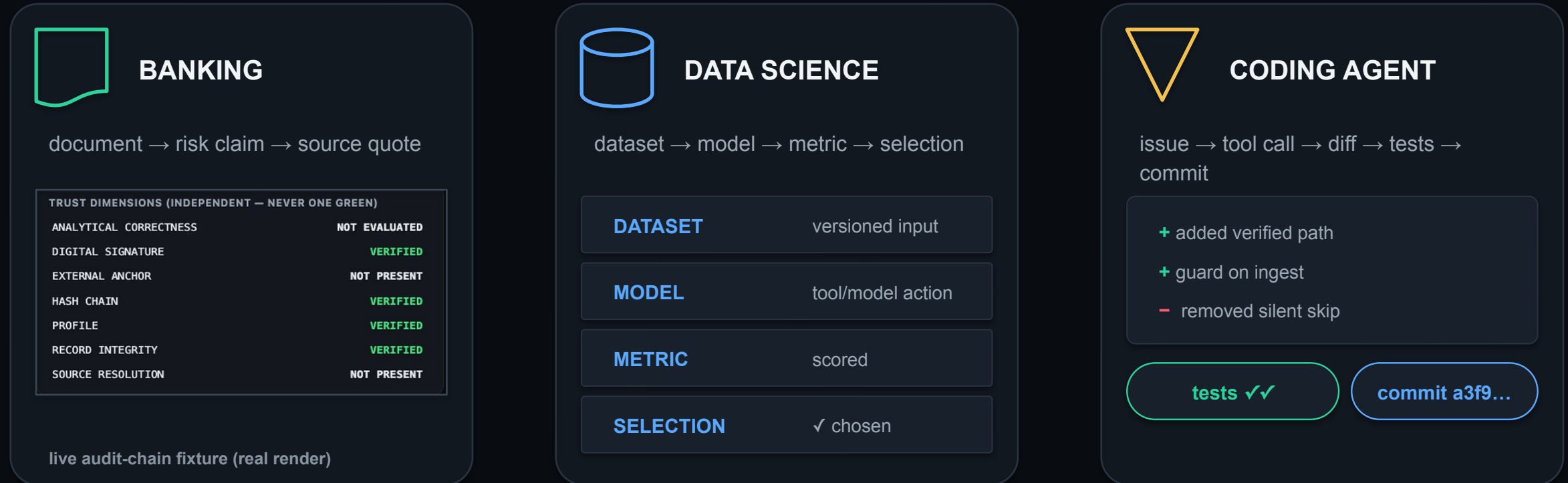
SHADOW CORE

canonical events · hash chain · Ed25519 · source maps · claim–evidence graph · ingest audit

HOST-TESTED

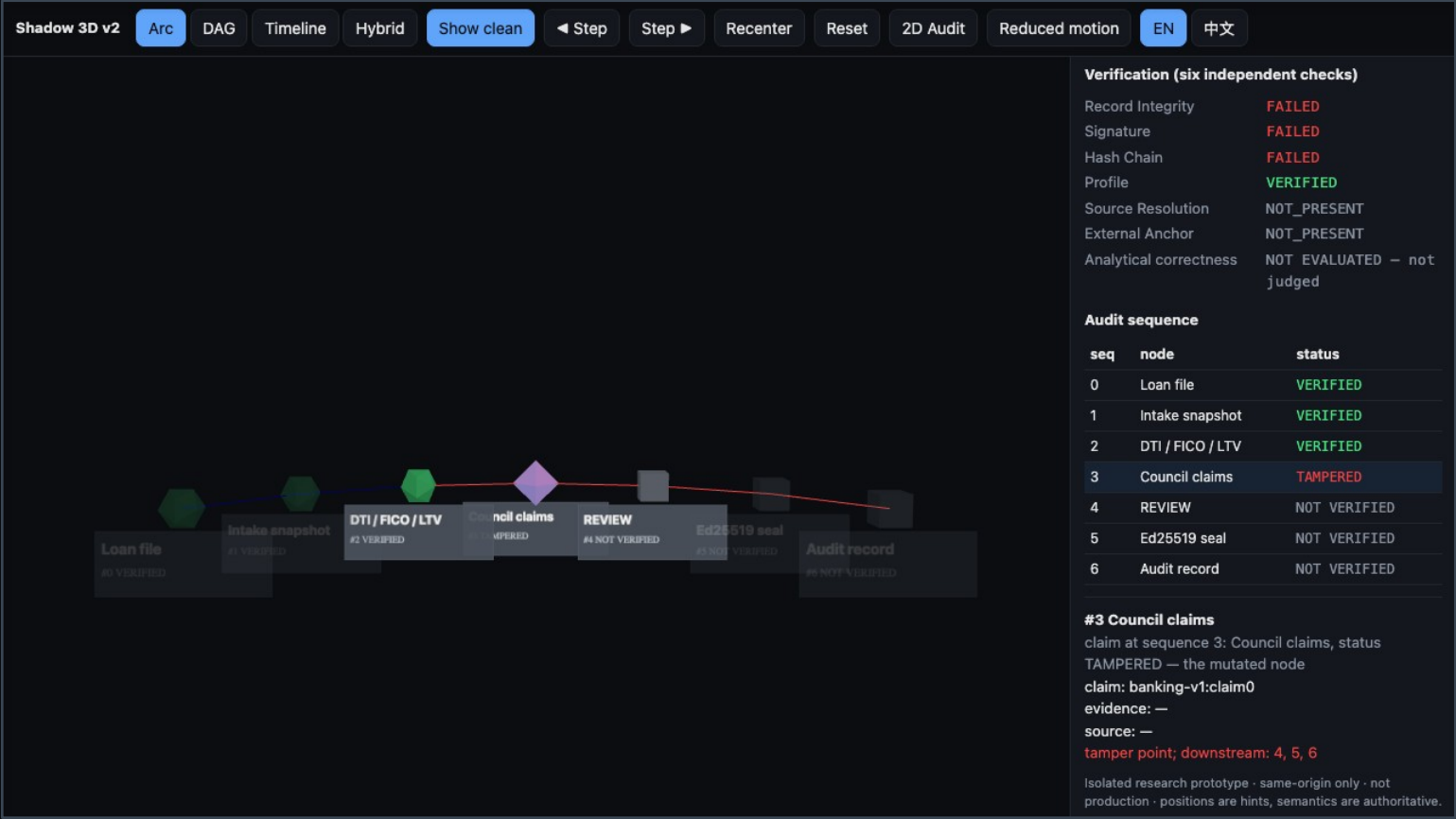
A portable **evidence bundle** flows through every layer — verifiable by anyone, offline. Interface maturity is labelled, not hidden.

Same machinery, domain-specific evidence



Underneath, identical: **sequence → hash chain → signature → source resolution**

Exact tamper localization



- 1 Seq 3 — TAMPERED
the mutated Council-claims node
- 2 Downstream 4, 5, 6
NOT VERIFIED — chain invalidated
- 3 Analytical correctness
NOT EVALUATED — not judged

Shadow reports:

first failed seq: 3

reason: prev_hash_mismatch

downstream: 4, 5, 6

Integrity failure does not determine analytical correctness — the two stay independent.

Can you trust the verifier page itself?

- 1 **Manifest signature**
VERIFIED
- 2 **Assets match**
SIGNED MANIFEST
- 3 **Independent comparison**
NOT PERFORMED

A page hashing itself is **not independent trust**.

Signing today is a **FIXTURE release key**, not production.

VERIFY EVIDENCE

VERIFY THE VERIFIER

Verifier integrity

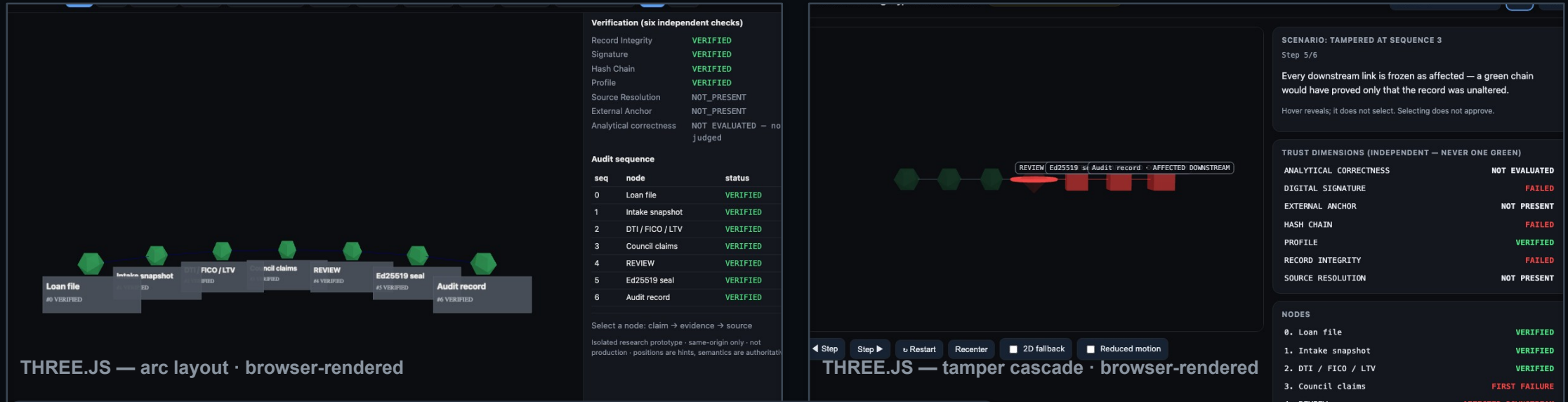
This page can verify that its loaded assets match a signed release manifest. That does not by itself prove that the manifest or release key was obtained from an independent trusted channel. For strongest assurance, compare the manifest fingerprint with the GitHub Release, npm package, or an approved offline copy.

Load release manifest

Execution mode	STATIC HOST
Release manifest signature	VERIFIED
Release signing-key fingerprint	727d29d3204231f7
Verifier integrity	ASSETS MATCH SIGNED MANIFEST
INDEPENDENT COMPARISON	INDEPENDENT COMPARISON NOT PERFORMED
NOT PERFORMED	

Real render — Verify-the-Verifier (fixture release key)

3D for sequence, provenance, tamper propagation



THREE.JS STORY ADAPTER host-tested contract + browser-rendered · 5 layouts + 2D fallback

SHARED SCENE / STORY CONTRACT authored + host-tested (shadow-3d-scene-v1)

UNITY SHADOW LENS unity-authored · C# contract-drift host-tested · Android-built

device-validation pending · not pictured (no Game View capture)

Honest bounds:

- not eye tracking
- no 6DoF claim on mock
- no user study yet (RQ4)

Measured today — not inferred

1,892

of 1,895 host tests

0 failed

3 skipped (env-gated)

36 suites

BROWSER

Chromium 149 · EN + 简体中文 · 0 external requests · 0 CSP violations · offline verify

ANDROID

APK built · 24.4 MB · SHA-256 recorded · NOT device-validated

SPATIAL

Scene + story contract host-tested · Three.js adapter host-tested + browser-rendered · Unity C# contract-drift tested · no user study

Re-run 2026-07-21 on feat/shadow-shared-story-adapters @ 5f655e8.

Accurate statuses, not blanket 'pending'

NOW

- ✓ Core evidence + hash-chain verification
- ✓ Claim–evidence graph (host-tested)
- ✓ Bilingual offline verifier · 3 explainers
- ✓ Shared scene + story contract — authored + host-tested
- ✓ Three.js story adapter — host-tested + browser-rendered
- ✓ Unity C# contract-drift tested · Android build

NEXT — CAPSTONE II

- Unity production integration + editmode/playmode
- Beam Pro device validation
- XREAL native input / camera path
- Device performance measurement
- Spatial-vs-2D user study (RQ4)
- Production signing · KMS/HSM · governance

Ingest audit: **structural host-tested** · **semantic production-evaluation pending** — not device-pending.

CONTRIBUTION

Shadow does not ask an auditor to trust the AI's answer.

It gives them evidence they can independently verify.

Sequence. Sources. Signatures. Exact tamper location.

Integrity — not correctness.

Capability ladder — nothing over-claimed

Evidence schema · hash chain · Ed25519

HOST-TESTED

Offline browser verifier (EN + zh-CN)

BROWSER-RENDERED / RECORDED

Verify-the-Verifier manifest

FIXTURE-SIGNED

Claim—evidence graph · 3 explainers

HOST-TESTED

Shared scene + guided-story contract

AUTHORED + HOST-TESTED

Three.js story adapter

HOST-TESTED · BROWSER-RENDERED

Unity Shadow Lens

UNITY-AUTHORED · C# DRIFT-TESTED · DEVICE-PENDING

Android mock APK

ANDROID-BUILT (not device-validated)

Ingest audit — structural / semantic

HOST-TESTED / PRODUCTION-PENDING

Spatial comprehension benefit (RQ4)

NOT IMPLEMENTED — no user study

What a valid signature does and does not prove

ESTABLISHES

- ✓ The bytes match the sealed record
- ✓ The chain is unbroken (or where it breaks)
- ✓ The batch root was signed by a known key
- ✓ Conformance to a domain profile

DOES NOT ESTABLISH

- ✗ That the cited source was truthful
- ✗ That the conclusion is correct
- ✗ That signing is production-grade
- ✗ That any regulation mandates it

Stated directly

- Evidence is fixture data; limited live-provider validation
- Production signing not implemented (fixture release key)
- Unity is authored + C# contract-drift-tested — not editmode/playmode-run, not device-validated
- Device validation pending (Beam Pro); XREAL native not integrated
- No completed user study — RQ4 spatial benefit unproven
- Semantic ingest audit: production evaluation pending (structural is host-tested)
- No proof of source truth; no proof of analytical correctness
- No production KMS/HSM, key rotation, or durable storage

Never depend on a live device

A · Live app

serve verifier package locally

B · Browser package

open the frozen offline package

C · Recorded video

Open bookmarked fallback video (Shadow verify full demo)

D · Slide screenshots

embedded figures — one keystroke away

Decision rule: if the live app doesn't come up cleanly in 20 seconds, open the bookmarked fallback video and keep talking.